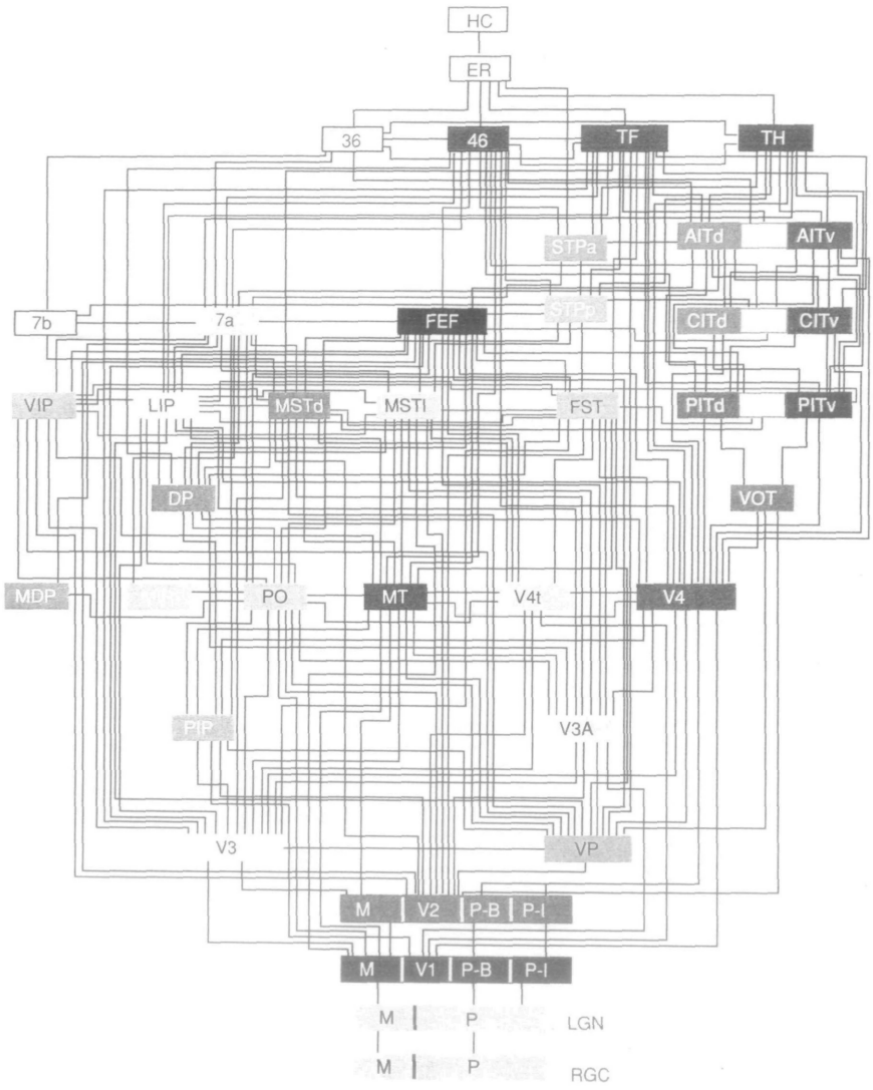
Outline

- HW1
- Single Neuron Model
- From Single Neuron to Multi Neuron

HW1

- Problem 1
  - cortical column
  - Octopus, Birds
- Problem 2
  - scatter vs meshgrid
  - axis equal
  - overlay
- Problem 3
  - The Visual System of Mammals
    - outside world -> cornea
    - cornea -> retina
    - retina -> V1
    - V1 -> ...



(figure from [Felleman et al., 1991])

Single Neuron Model

- Hopfield Model
- M-P Model
- Lapique Model
- HH Model

From Single Neuron to Multi Neuron

- From V to r
  - count spikes
  - Tuning Curve
- FNN
- RNN
- CNN
  - [Yann LeCun, CNN, 1998]

- AlexNet, CNN, 2012